

CHICHIBU, Japan

WMO: 476410

Lat: 35.9900N

Lon: 139.0742E

Elev: 767

StdP: 14.29

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.6	23.5	5.5	7.5	40.4	8.6	8.8	39.5	15.6	42.9	14.0	43.3	2.1	220	0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.9	94.0	77.8	91.3	77.0	88.4	75.7	79.3	90.0	78.2	88.2	77.1	85.7	6.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.5	142.2	83.9	75.5	137.6	82.7	74.6	133.4	81.7	43.5	90.3	42.4	88.7	41.3	85.9	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.9	11.2	9.6	18.1	98.0	1.3	1.7	17.2	99.3	16.4	100.3	15.6	101.3	14.7	102.5
			16.9	81.4	1.3	1.0	15.9	82.1	15.1	82.7	14.4	83.3	13.4	84.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.3	36.3	38.1	44.3	54.5	63.8	69.9	77.4	79.3	71.6	61.0	49.6	40.5
	DBStd	15.68	3.66	4.58	5.70	6.40	5.11	4.84	5.33	4.69	5.66	5.37	5.33	4.60
	HDD50	1353	425	335	196	28	0	0	0	0	0	1	72	296
	HDD65	4076	890	754	641	319	83	12	0	0	10	148	461	758
	CDD50	4019	0	1	20	163	427	598	850	907	648	341	61	3
	CDD65	1267	0	0	0	3	45	160	385	442	209	23	0	0
	CDH74	10152	0	0	1	79	497	1036	3255	3916	1264	104	0	0
	CDH80	3872	0	0	0	12	102	285	1396	1680	382	15	0	0
(14) Wind	WSAvg	3.4	3.6	3.9	4.0	4.0	3.7	3.3	3.2	3.2	2.9	2.6	2.7	3.2
(15) Precipitation	PrecAvg	56.1	1.8	1.6	3.0	3.8	4.8	6.4	6.9	7.2	9.0	7.0	2.8	1.8
	PrecMax	80.2	4.8	3.6	5.6	11.3	10.6	17.4	13.6	19.1	17.5	21.6	12.5	5.6
	PrecMin	36.3	0.2	0.2	1.0	1.1	1.9	2.4	2.6	1.2	1.6	0.9	0.2	0.3
	PrecStd	9.0	1.4	0.9	1.2	2.4	2.2	2.9	3.1	4.7	5.0	5.0	2.3	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	63.5	71.6	80.9	87.2	91.1	96.8	97.0	91.4	82.0	70.2	61.6
		MCWB	44.5	50.0	54.4	61.4	66.5	73.6	78.3	78.3	76.2	69.3	57.8	49.1
	2%	DB	52.4	56.5	65.5	75.7	82.7	86.3	94.1	94.3	87.7	76.6	66.5	57.3
		MCWB	40.7	44.2	50.5	57.8	64.0	70.7	78.0	77.9	74.7	65.3	55.6	46.6
	5%	DB	49.2	52.4	61.1	71.8	79.4	83.2	91.2	91.9	84.4	73.3	63.3	54.0
		MCWB	38.4	41.1	48.0	56.4	62.8	69.3	77.2	77.6	73.4	63.4	53.0	44.4
	10%	DB	46.6	49.0	56.7	68.0	76.2	80.1	88.1	89.2	81.3	70.2	60.0	50.9
		MCWB	36.8	39.0	46.0	54.7	61.6	68.2	76.1	76.8	71.6	61.9	52.2	42.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.8	51.7	56.3	65.2	70.8	76.9	80.7	81.0	78.2	71.5	61.3	52.9
		MCDB	52.7	61.6	66.9	73.9	80.4	86.5	91.9	92.2	87.5	79.0	66.3	57.4
	2%	WB	42.4	46.0	52.9	61.5	67.7	74.3	79.2	79.6	76.5	68.4	58.5	48.7
		MCDB	49.5	54.1	61.2	71.6	77.3	81.9	90.2	90.4	84.3	73.4	62.6	54.2
	5%	WB	40.0	42.9	50.1	58.8	65.6	72.3	78.2	78.6	75.1	66.1	56.0	46.0
		MCDB	47.1	49.9	58.5	66.6	74.2	78.9	88.2	88.9	81.6	70.4	60.3	51.9
	10%	WB	37.7	40.2	47.3	56.6	63.9	70.5	77.0	77.6	73.5	64.0	53.8	43.7
		MCDB	45.0	47.1	54.8	64.6	72.1	76.4	85.7	86.5	79.0	67.9	58.0	49.1
(35) Mean Daily Temperature Range	5% DB	MDBR	21.7	21.5	21.5	22.1	20.3	15.8	15.4	15.9	14.6	16.1	20.0	21.6
		MCDBR	26.0	27.5	28.9	28.8	27.8	22.9	21.7	21.3	19.5	20.4	24.0	25.2
		MCWBR	15.5	16.3	15.9	13.7	12.1	9.3	8.1	7.8	8.2	9.9	13.4	14.9
	5% WB	MCDBR	24.4	24.9	25.4	23.3	22.1	17.6	19.6	18.9	16.8	16.4	19.1	22.3
		MCWBR	15.5	15.6	15.3	12.6	11.1	8.0	8.0	7.6	7.6	8.8	11.7	14.1
(40) Clear-Sky Solar Irradiance	taub		0.356	0.402	0.467	0.522	0.555	0.599	0.588	0.589	0.513	0.466	0.397	0.357
	taud		2.253	2.120	1.976	1.892	1.868	1.827	1.910	1.897	2.062	2.113	2.240	2.304
	Ebn at Noon		260	260	252	245	239	228	230	226	237	236	243	252
	Edh at Noon		33	43	54	62	64	67	61	61	49	43	34	30
(44) All-Sky Solar Radiation	RadAvg		845	1047	1309	1525	1645	1459	1500	1503	1202	988	812	744
	RadStd		49	81	81	138	168	119	231	146	105	101	69	54

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page